

Exchange rate and Production network

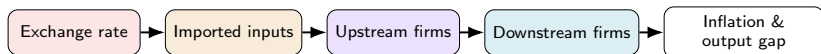
Bonn Macro Internal Seminar

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Motivation



Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

Does accounting for production networks change the optimal exchange-rate regime?

Literature

Network side (closed economy): Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

Open-economy side (no network): Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

Open economy with network:

Gnocoato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025) — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice¹

Qiu, Wang, Xu & Zanetti (2026, JME) — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

Gap: network + open economy + FX regime choice, together.

¹Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo (2024) → SOE: imported inputs, UIP and debt-elastic premium

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Calibrate: Chile's real IO table, also Korea, Czechia

Ask: rank FX regimes? How does the network amplify foreign shocks?
How do sector heterogeneity and network structure matter?

Result: Managed < Float $\ll$ Peg (welfare loss
 $10.96/25.15/90.45 \times 10^{-4}$)

Households

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize lifetime utility, subject to a budget constraint with a domestic bond B_t and a foreign bond B_t^* :

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1})B_{t-1} + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)]S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition:

$$Y_{ift} = A_{it} L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H} M_{ift}^{\Omega_i^F}$$
$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) and S_t , the exchange rate (via the import cost $S_t P_t^{F*}$).

The theory holds for any number of sectors N ; the quantitative results use $N = 3$ (Resource, Manufacturing, Services), collapsed from Chile's 12-sector national IO table.

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H \Delta)^{-1}$, applied to labor share α vs. import share ω_F :

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa}, \quad \Gamma = \frac{\tilde{A}\omega_F}{\kappa}$$

where:

$\pi_t, \tilde{y}_t$: infl., gap	χ_t : TFP cost-push	Δe_t : exch. rate chg.	κ : GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Firms: Production Network

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \dots$$

applied to consumption shares β^H (Households) it gives:

$$\lambda_D^\top = (\beta^H)^\top (I - \Omega^H)^{-1}, \quad \mathcal{M} = (I - \Omega^H)^{-1} \omega_F, \quad \mathcal{M}^X = (I - \Omega^H)^{-1} \omega_X$$

where:

$\lambda_{D,i}$: **Domar weight** — sector i 's total weight in final demand, network-adjusted (can exceed 1 for downstream sectors)

$\mathcal{M}_i$: **Import centrality** — sector i 's total (direct + indirect) exposure to import costs

$\mathcal{M}_i^X$: **Export centrality** — sector i 's total (direct + indirect) exposure to export demand

Foreign Sector

Law of one price and UIP with a debt-elastic NFA stabilizer $\psi(\cdot)$ and an exogenous risk-premium shock RP_t :

$$P_t^{FH} = S_t P_t^{F*}, \quad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance:

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \quad EX_{it} = \kappa_{EX,i} D_t^* PX_t \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \rightarrow MC_{it} \rightarrow \pi_{it} \quad (\text{cost-push}) \quad \hat{s}_t \rightarrow EX_{it} \rightarrow \tilde{y}_t \rightarrow \pi_{it} \quad (\text{demand})$$

where:

P_t^{F*} : foreign price of imports
 $\psi(\cdot)$: debt-elastic NFA stabilizer
 $\hat{s}_t \equiv \log(S_t/\bar{S})$: log exch. rate

RP_t : risk-premium shock
 $\hat{b}_t^*$: NFA (% GDP)
 $\Delta e_{t+1} \equiv \mathbb{E}_t \hat{s}_{t+1} - \hat{s}_t$: exp. depreciation

D_t^* : foreign demand shock
 $\kappa_{EX,i}$: export scale, sector i
 PX_t : export price/ToT shock
 θ^* : export demand elasticity
 (hats: log-deviation from steady state, throughout)

Monetary Policy Regimes

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Phillips curve:

$$\pi_t = \rho(I - \mathcal{V})\mathbb{E}_t\pi_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\chi_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t\pi_t^C - r_t^{\text{nat}}) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t & \text{Managed} \end{cases}$$

Seven AR(1) shocks (1% s.d. each): sector TFP A_1, A_2, A_3 ($\rho=0.90$), import price PF ($\rho=0.85$), foreign demand D^* ($\rho=0.80$), export price/ToT PX ($\rho=0.85$), risk premium RP ($\rho=0.80$).

Calibration: Chile Data

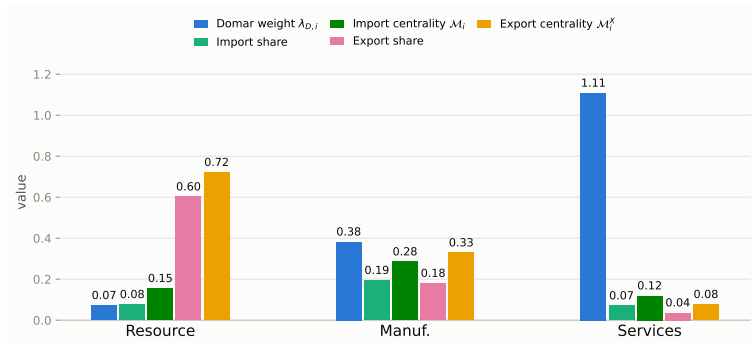
Param.	Description	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	Labor share	0.503, 0.359, 0.604	Data: BCCh IO
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	Import cost share	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	HH consumption share	0.027, 0.229, 0.744	
Export share _{1,2,3}	Export share of output	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	Calvo reset prob.	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); N-S (2008)
ψ	Risk-premium elasticity	0.020	Key mechanism param. (see appendix)
ϕ_π	Taylor: inflation resp.	1.50	
ϕ_y	Taylor: output-gap resp.	0.50	
ϕ_s	Taylor: FX-smoothing resp.	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

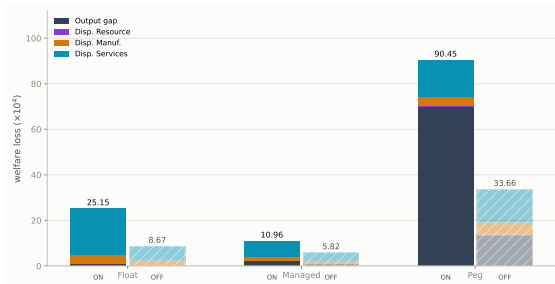
$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): $\beta=0.99$, $\gamma=1.00$, $\varphi=2.00$, $\eta=1.50$, $\varepsilon=8.00$, $\theta^*=2.00$, $\beta^F, \text{tot}=0.10$.

Network Properties: Import & Export Exposure



Welfare Ranking: Managed Float Dominates



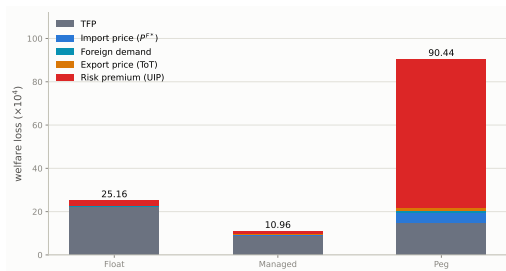
Managed < Float < Peg: $10.96/25.15/90.45 (\times 10^{-4})$

Float & Managed: almost entirely price dispersion, concentrated in Services

Peg: almost entirely output gap

Without the network: same ranking, but the network roughly **doubles-to-triples** every regime's loss

What Drives Each Regime's Loss? Shock Decomposition



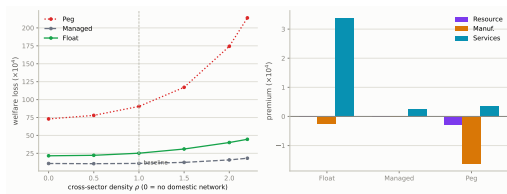
Peg: risk-premium/UIP $\rightarrow$ 76% of loss, mostly output gap

Float & Managed: TFP shocks dominate instead — risk-premium barely registers since the exchange rate (Float) or partial smoothing (Managed) absorbs it before it reaches output

Import price, foreign demand, and export price are second-order for every regime

Isolating the Network Channel

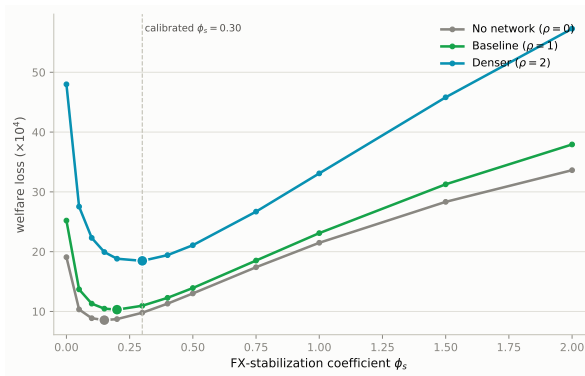
Does the network's effect scale *continuously* with how interconnected the economy is?



Denser network amplifies shock propagation for *everyone*, but **Peg** has no policy lever ($\phi_\pi, \phi_y = 0$) to lean against it, so the amplification passes straight into welfare loss; **Float/Managed** actively damp part of it with the interest rate — **gap widens** as $\rho \uparrow$

Does the Optimal Managed-Float Response Depend on the Network?

Christian's question: welfare loss changes a lot with network density — does the **welfare-minimizing** FX-stabilization coefficient ϕ_s change too, or just its magnitude?



Re-solves the managed-float rule $\hat{t}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log S_t$ over the ϕ_s grid at three network densities: **no network** ($\rho=0$), **baseline** ($\rho=1$), **denser** ($\rho=2$)

ϕ_s^* **rises with density**: 0.15 $\rightarrow$ 0.20 $\rightarrow$ 0.30 — denser network, more indirect cost-push (previous slide), so the welfare-minimizing rule leans harder against the exchange rate

How the Network Amplifies Shocks

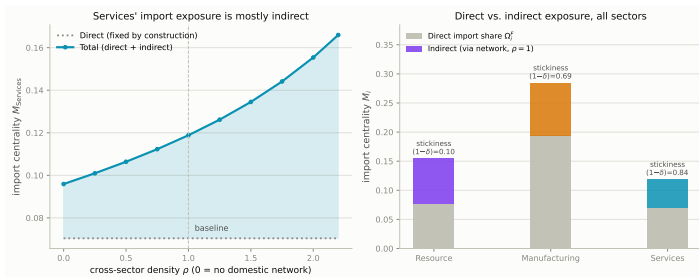
The network **dampens** the inflation peak but **amplifies** the output-gap peak — in every regime, for *both* shocks.

Shock	Regime	Inflation peak (Network ON ÷ Network OFF)	Output-gap peak
Import-price	Float	0.40×	2.77×
	Managed	0.61×	2.28×
	Peg	0.86×	3.15×
Risk-premium	Float	0.41×	2.57×
	Managed	0.57×	0.98×
	Peg	0.67×	2.16×

Note: $< 1\times$ = network shrinks the peak; $> 1\times$ = network amplifies it. (Peak IRF response over 20 quarters; exchange-rate peak itself barely moves with the network)

Why Services? Direct vs. Indirect Import Exposure

Services is the **stickiest** sector (Calvo reset prob. $\delta_3=0.16$) and has the **lowest direct** import share ($\Omega_3^F=0.070$) — yet it buys 5.8% of its cost from Manufacturing, the most import-intensive sector ($\Omega_2^F=0.195$).



Total import centrality $M_i = [(I - \Omega^H)^{-1} \Omega^F \mathbf{1}]_i$ splits into a **direct** part (own Ω_i^F) and an **indirect** part inherited from suppliers — for Services, **41% of total exposure is indirect** (Resource 50%, Manuf. 32%)

Indirect exposure is **mechanically zero** at $\rho=0$ and grows with network density ρ — density doesn't change any sector's *direct* import share, only how much of *other* sectors' import content gets passed to it

Combined with stickiness: Services can't reprice quickly, so whatever cost-push it inherits shows up as persistent inflation, not a one-quarter blip — $\text{Corr}(\pi_3, \pi_3^{t-1}) \approx 0.82$, the most persistent inflation series in the model

Network Exposure & Sector Welfare Preferences

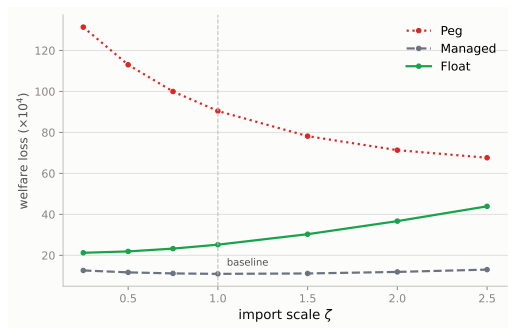
Each cell: **Network ON** / **Network OFF**.

Price-dispersion cost ($\times 10^4$)			
Sector	Float	Managed	Peg
Resource	0.05 / 0.03	0.04 / 0.02	0.27 / 0.27
Manuf.	3.87 / 2.23	1.92 / 0.94	4.28 / 5.32
Services	20.25 / 6.22	6.71 / 3.74	15.88 / 14.59

Var(own output gap) ($\times 10^4$)			
Sector	Float	Managed	Peg
Resource	0.13 / 0.02	0.14 / 0.04	0.20 / 0.12
Manuf.	0.58 / 0.16	0.38 / 0.08	1.98 / 0.23
Services	0.38 / 0.15	2.31 / 1.33	35.16 / 8.97

Every sector agrees: $\text{Managed} < \text{Float} < \text{Peg}$ — Services' huge stake in avoiding Peg is mostly **network-driven** (output-gap cost $8.97 \rightarrow 35.16$ with the network on).

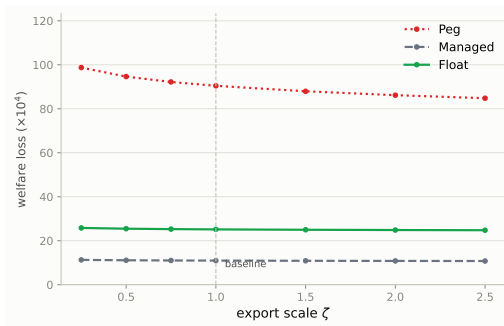
Import Openness



Managed < Float < Peg robust throughout

More import exposure means more $\Gamma \Delta e_t$ cost-push for **Float** (loss rises), but that channel is structurally zero under **Peg** ($\hat{s}_t \equiv 0$), so its loss instead falls via the labor-share renormalization — **gap converges**

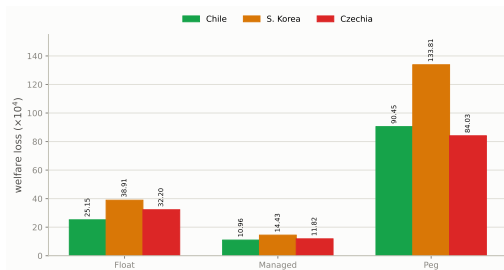
Export Openness



Managed < Float < Peg robust throughout

Exports work only through **demand** ($EX_{it} \rightarrow \tilde{y}_t$), never marginal cost — a pure buffer, so **all regimes improve**, and **Peg gains most** since it has no FX buffer of its own

Robustness: South Korea & Czechia



Managed < Float $\ll$ Peg in every calibration

Korea higher throughout: denser network & higher import exposure

Conclusion

The network matters for severity and incidence, not the ranking: switching it on raises welfare losses in every regime and reallocates who bears them, but **Managed < Float < Peg either way**

Managed float wins because it dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

Robust across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution